



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

**ENEE4113 - COMMUNICATION LAB Experiment
#11 PRELAB**

Frequency and Phase Shift keying

Prepared By:

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Instructor: Dr.Alhareth Zayoud

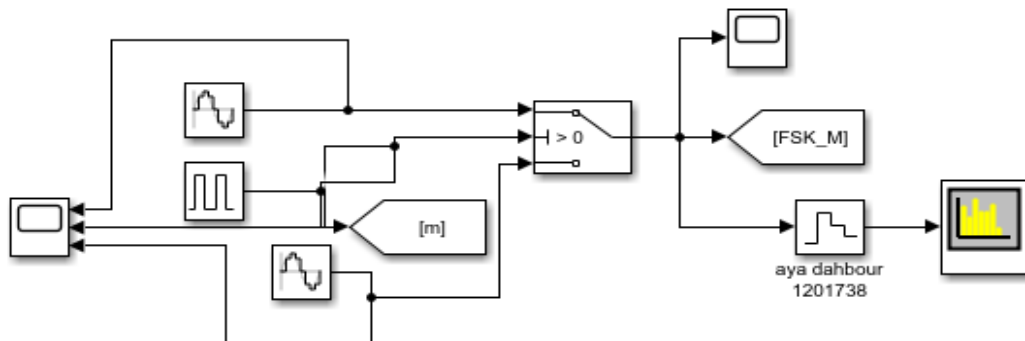
Assistant: Eng.Shadi Bannoura

Section: 4

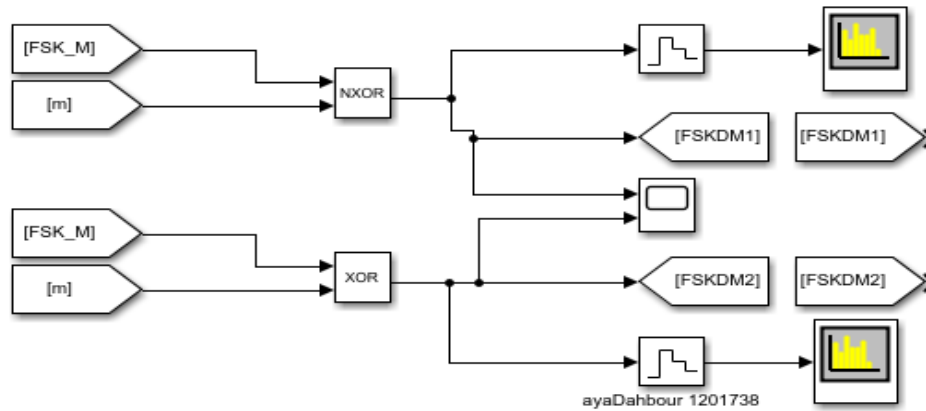
Date: 25/6/2023

Part1 : Blocks Diagram using MATLAB Simulink

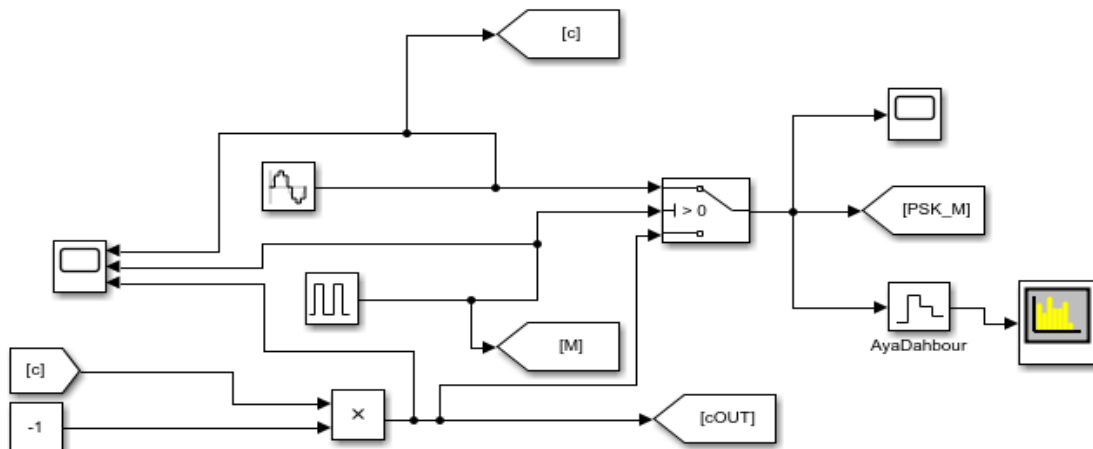
- Frequency Shift Keying Modulation Block diagram in the figure below.



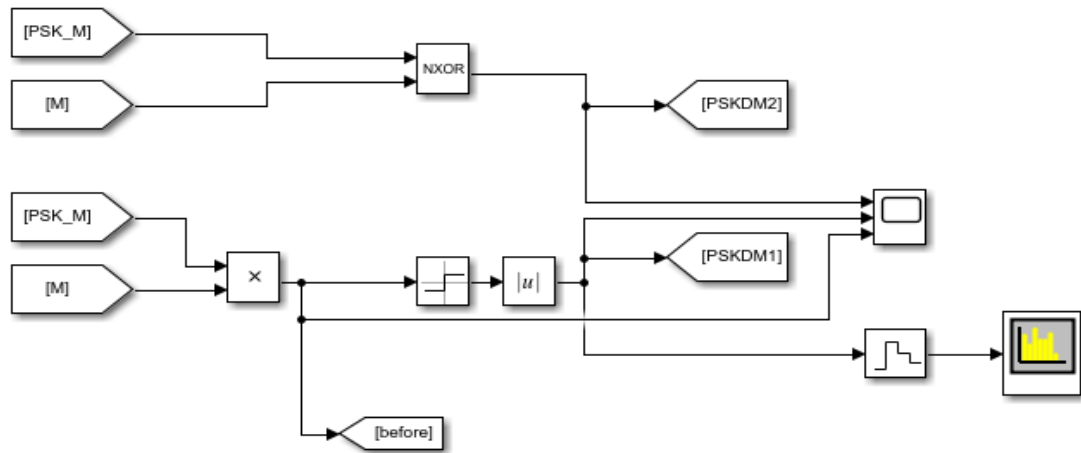
- Frequency shift keying Demodulation Block diagram in the figure below.



- Phase shift keying Modulation Block diagram in the figure below.



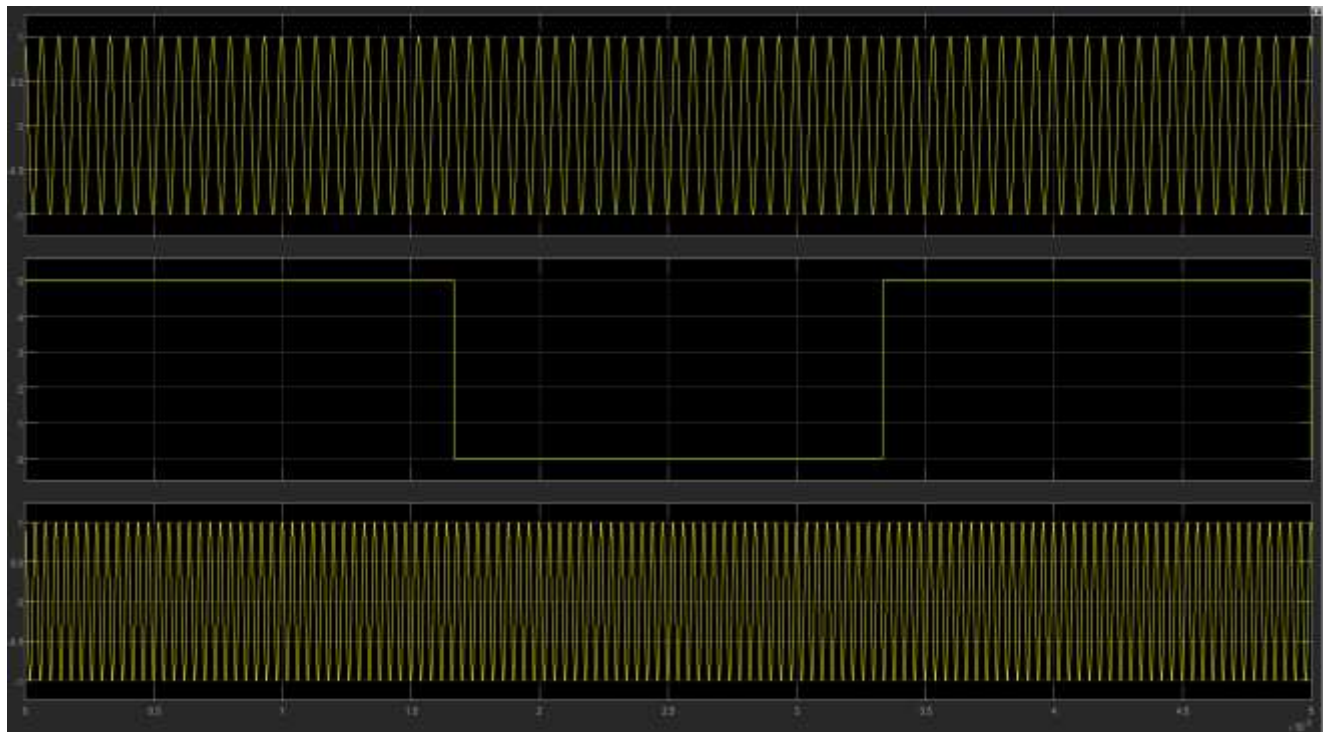
- Phase shift keying Demodulation Block diagram in the figure below.



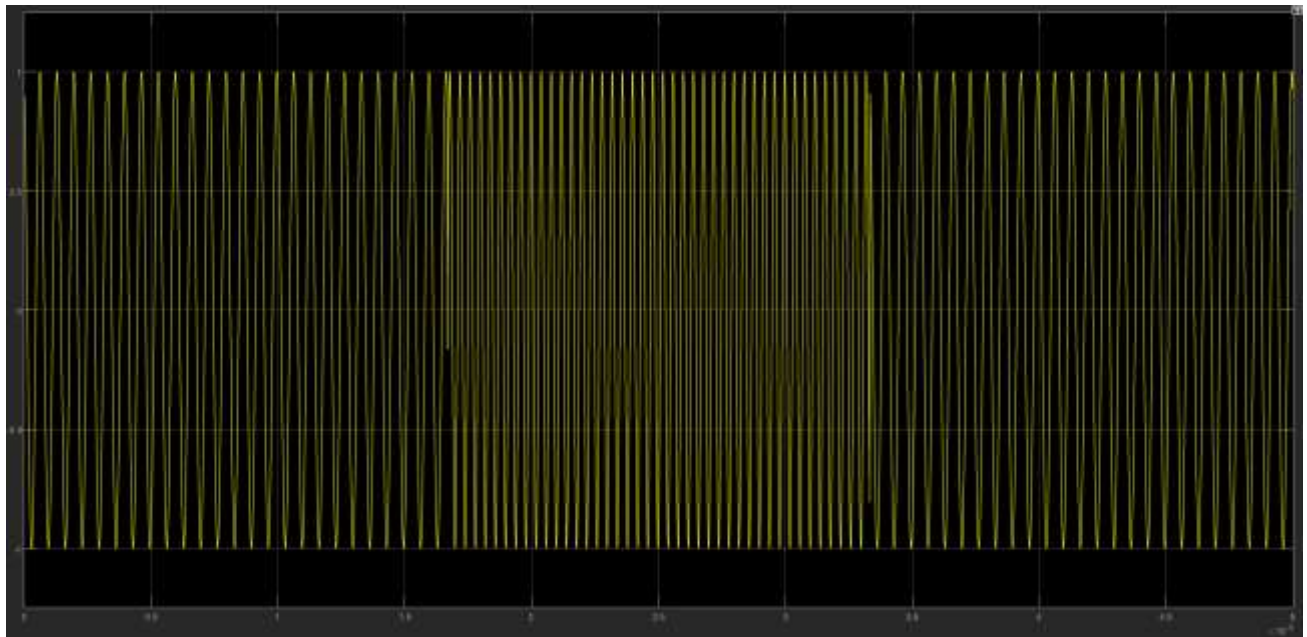
Part 2: The Results

- **Frequency Shift Keying Modulation**

- In time domain
(message pulse wave and 2 carrier)



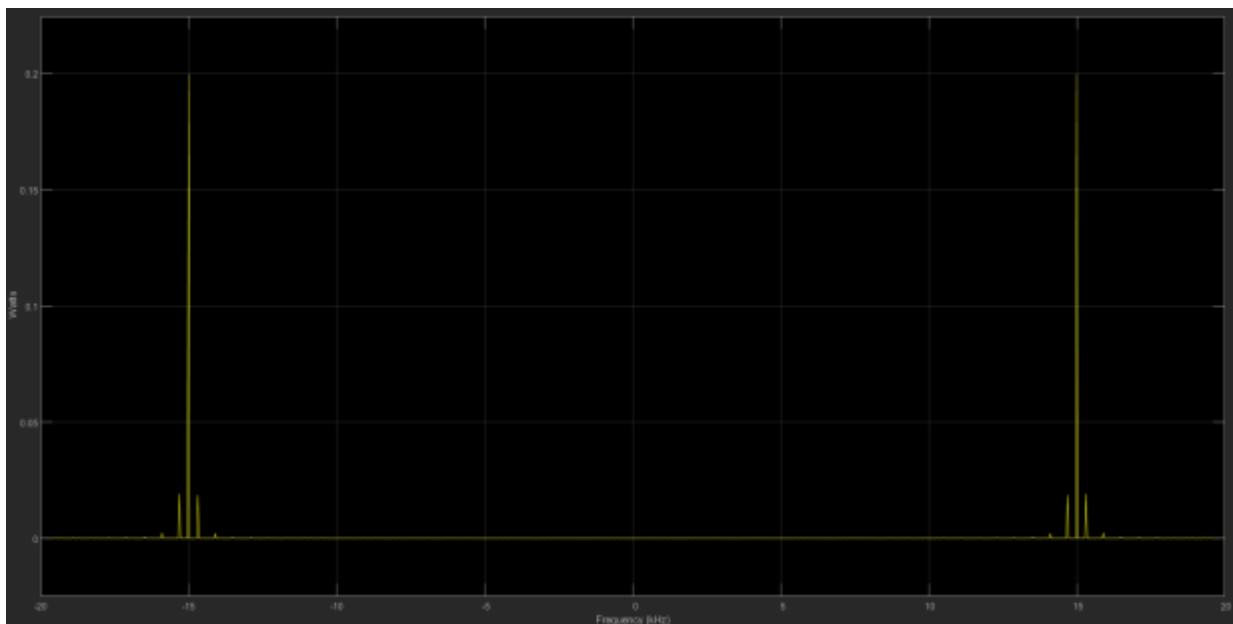
(FSK Modulated Signal)



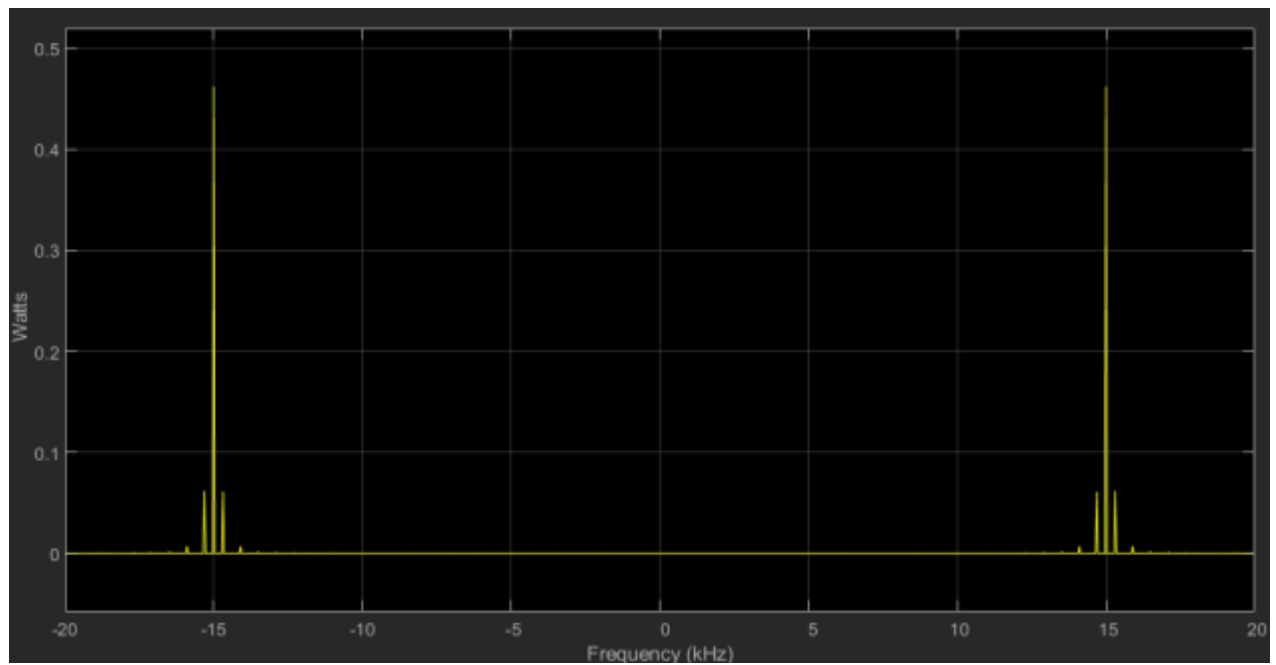
The relationship between message amplitude and modulated signal frequency is Direct proportion.

➤ In frequency domain

(the Modulated Signal)

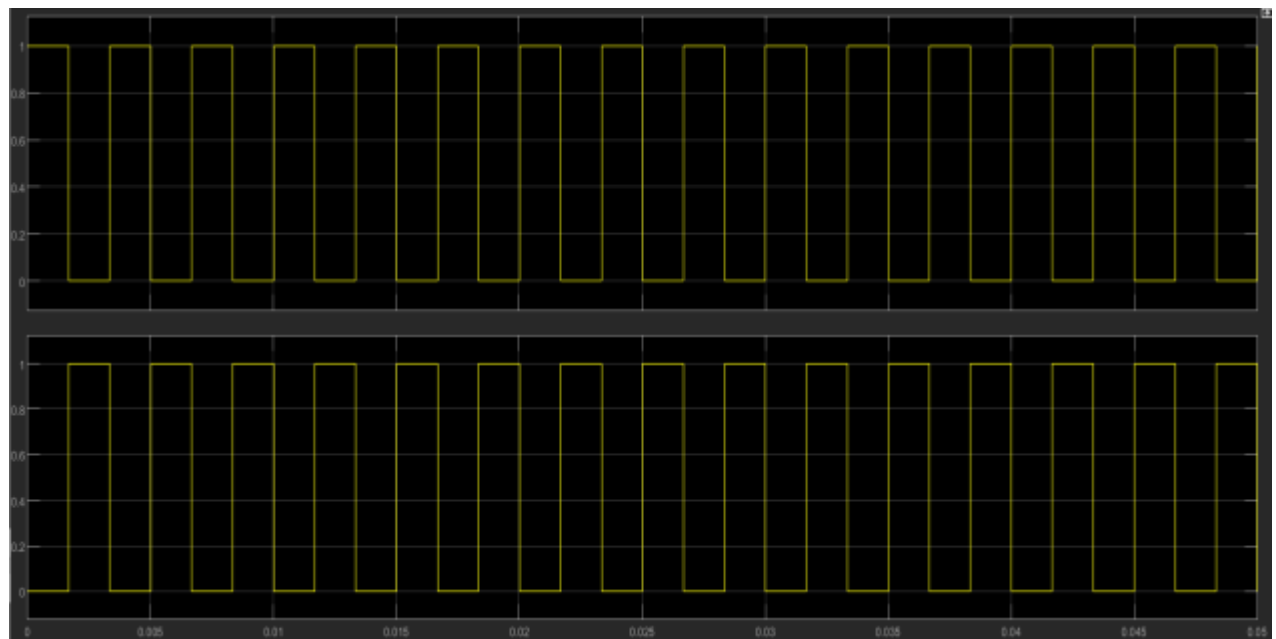


(After change the message amplitude to 2v)

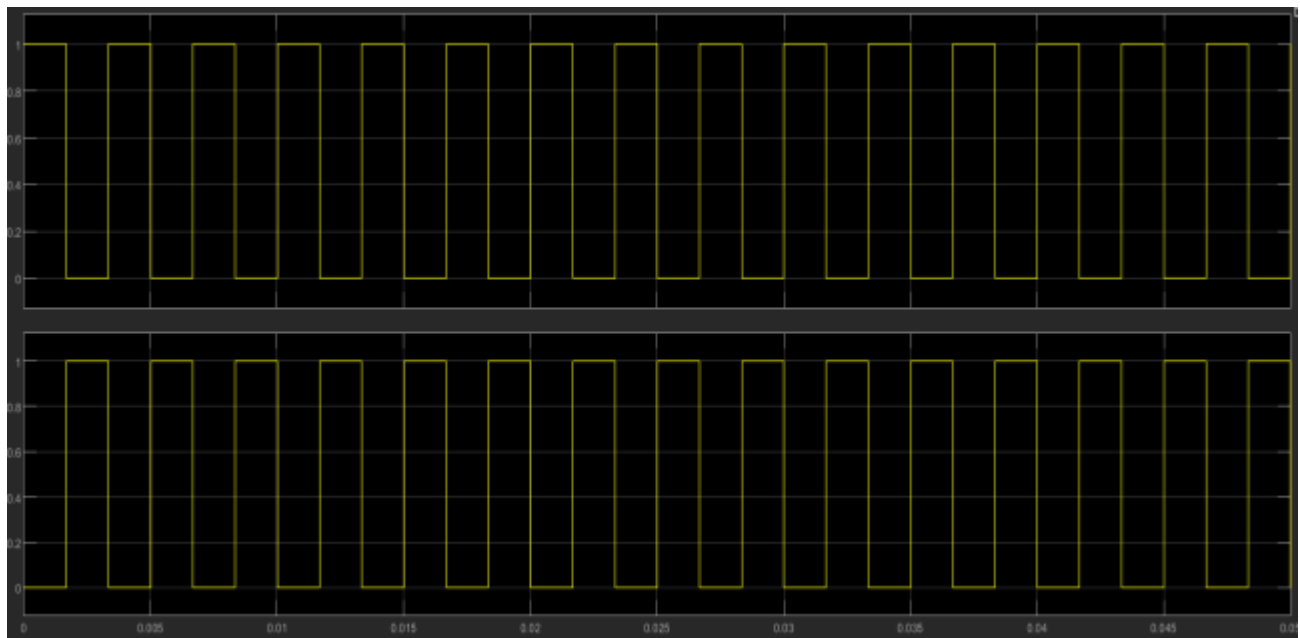


- **Frequency shift keying Demodulation**

- In time domain
($V=5$, $f=1\text{KHz}$ and 50%duty cycle)

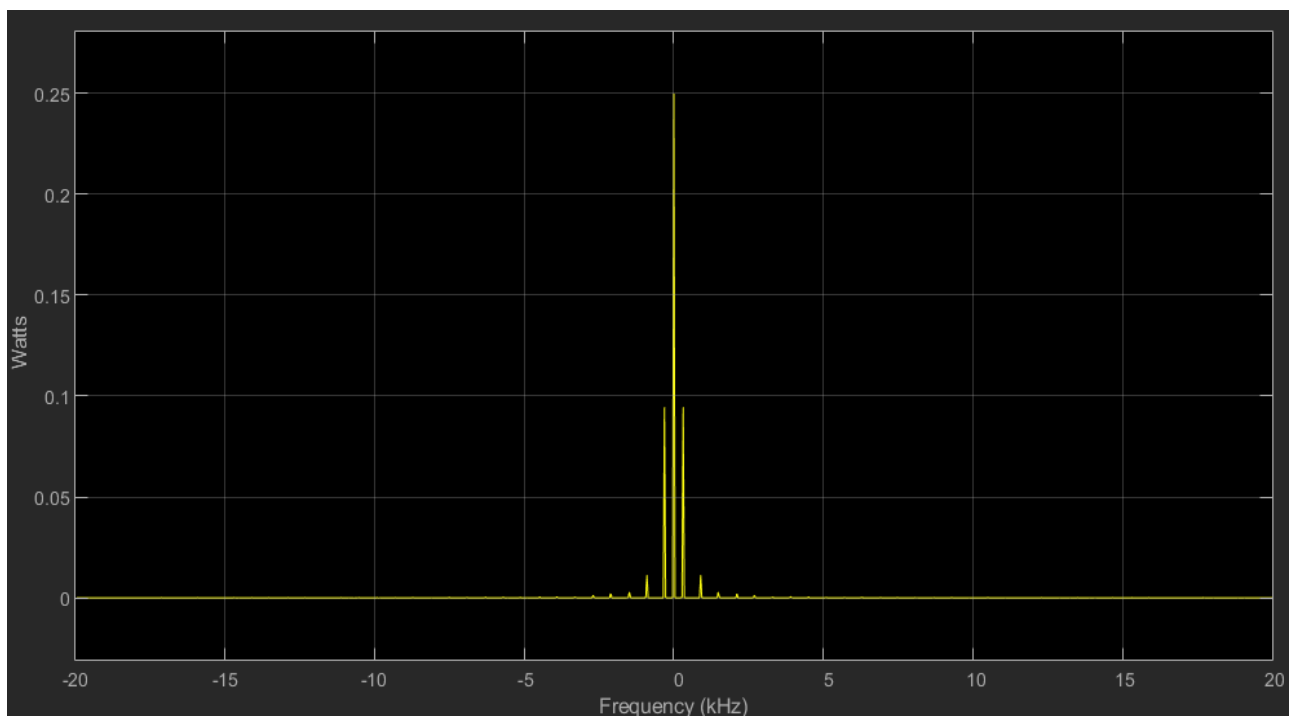


(After change the message frequency to 500Hz)



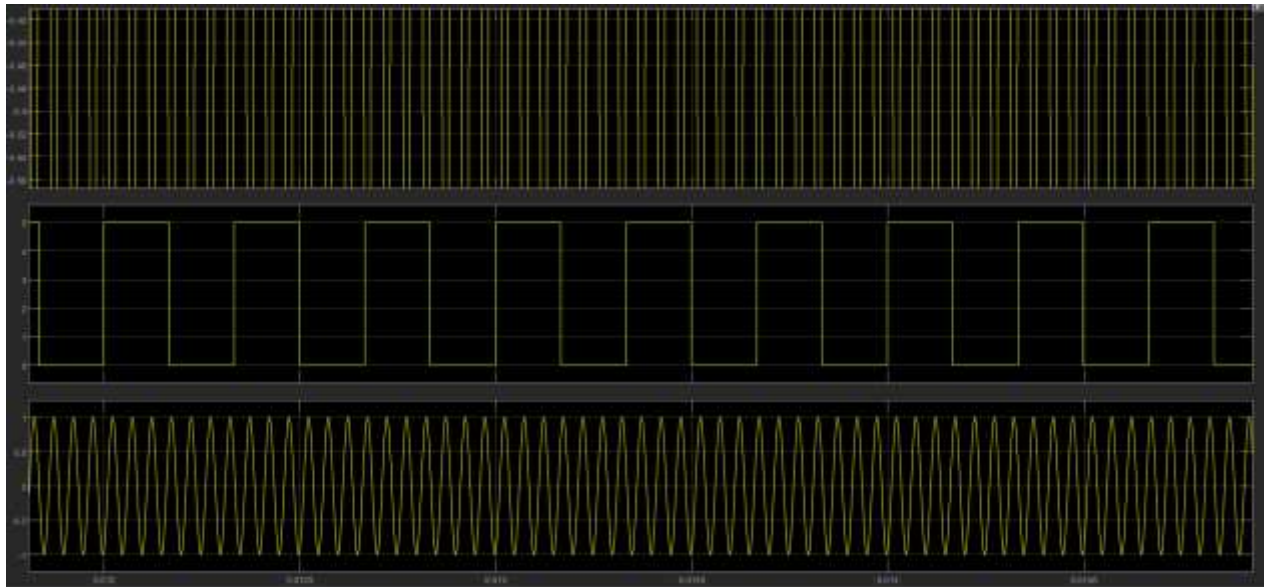
The periodic time grew when the frequency was reduced. That inverse proportion.

➤ In frequency domain

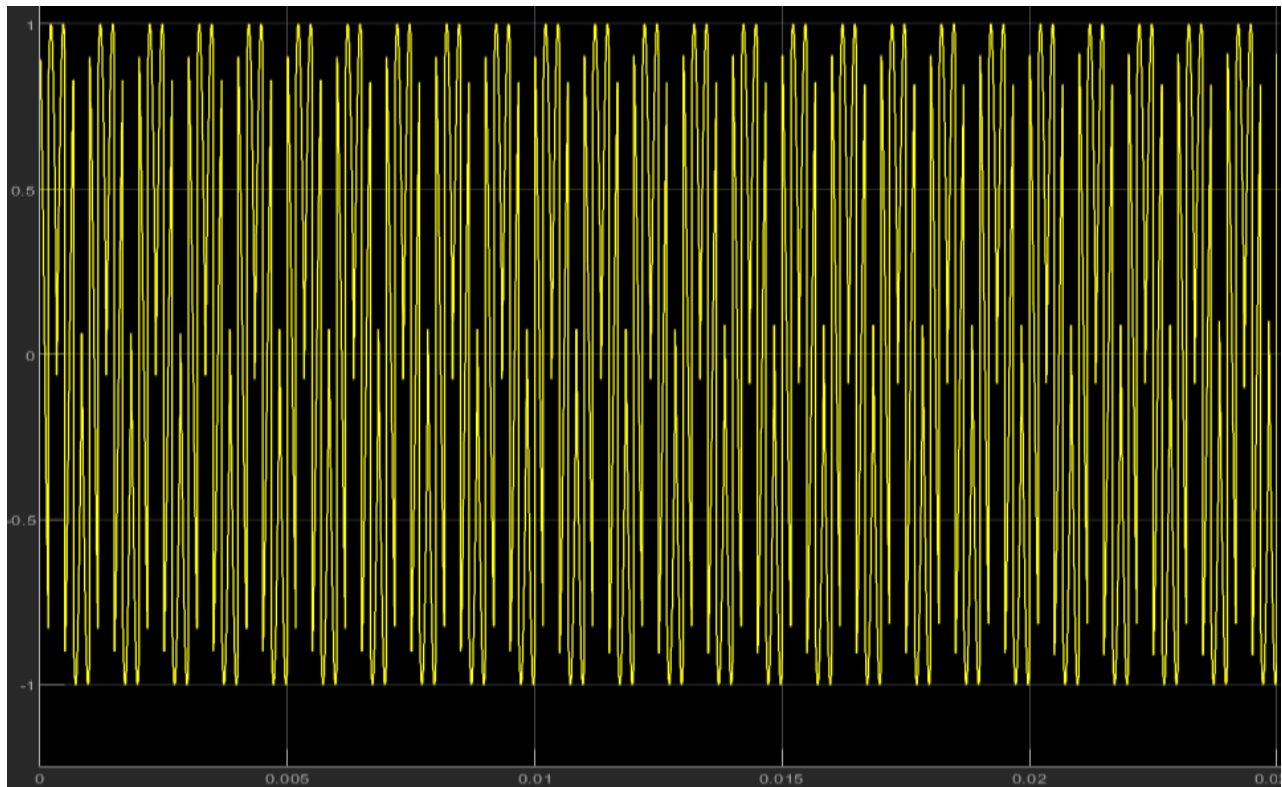


the demodulation done successfully.

- **Phase shift keying Modulation**
- In time domain
(message pulse wave and 2 carrier)

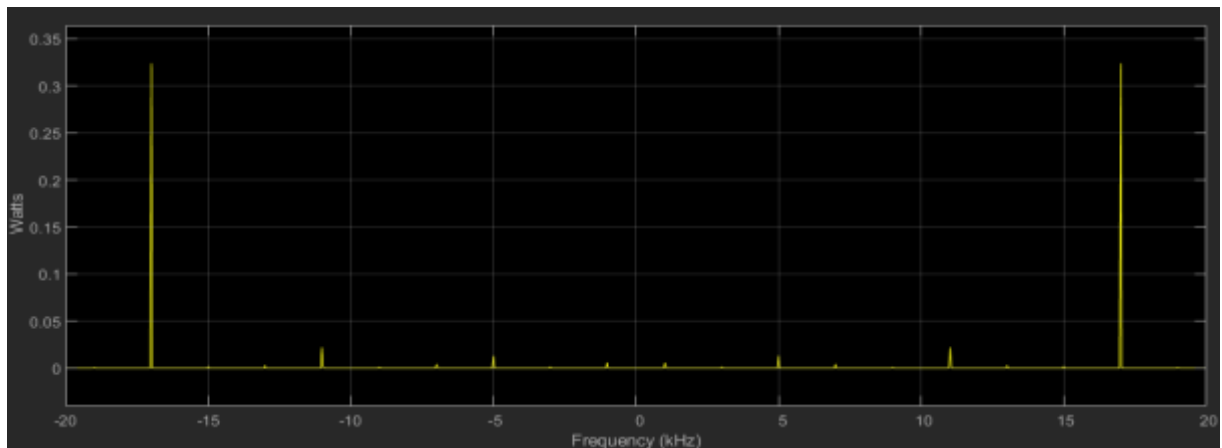


(PSK Modulated Signal)



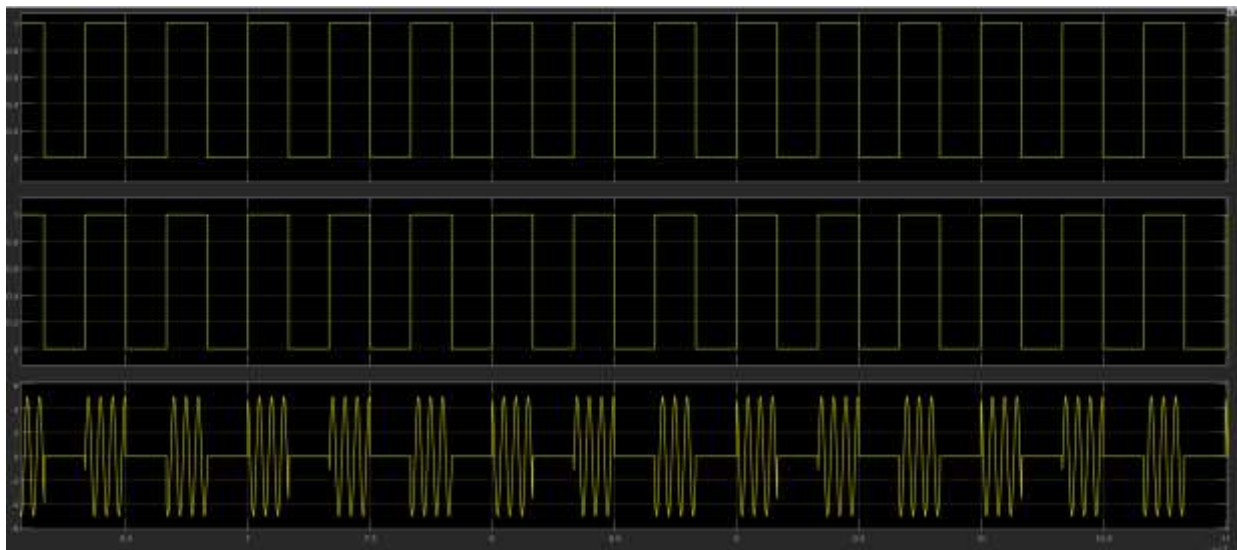
Wallah I don't know what happened here but this is the output.

- In frequency time

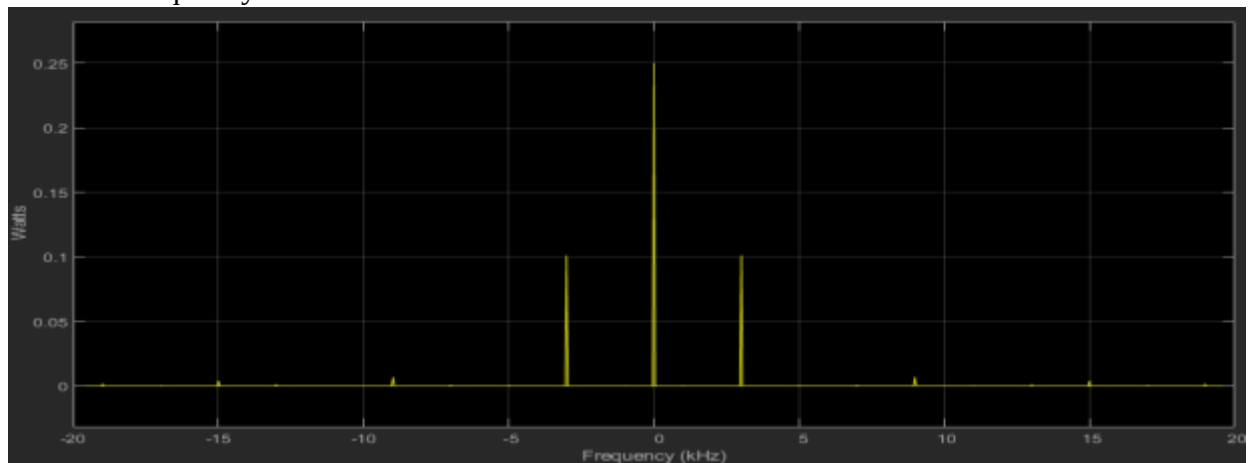


- **Phase shift keying Demodulation**

- In time domain



- In frequency time



Sitting:

